AT A MINIMAL PRIME EXACTLY ONE COHOMOLOGY MODULE OF A DUALIZING COMPLEX IS NONZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-dualizing/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The localization of a dualizing complex at a prime is a dualizing complex over the local ring. Invoked at the start of the counted proof (source lines 3456-3458). Transports the hypothesis to the local ring at the minimal prime.
- The shape of a dualizing complex over an Artinian local ring. Invoked at the end of the counted proof (source line 3460). Gives that a dualizing complex over a zero dimensional local ring has exactly one nonzero cohomology module.

Source reference. Original lemma lemma-dualizing-artinian: sources/stacks-dualizing/pinned-original-source.tex, source lines 3351–3351. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing-artinian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-dualizing-localize: sources/stacks-dualizing/pinned-original-source.tex, source lines 2950–2950. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing-localize, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring. Let $\mathfrak{p}$ be a minimal prime of A. Then $H^i(\omega_A^\bullet)_\mathfrak{p}$ is nonzero for exactly one i.*

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The complex $\omega_A^\bullet \otimes_A A_\mathfrak{p}$ is a dualizing complex for $A_\mathfrak{p}$ (Lemma lemma-dualizing-localize). The dimension of $A_\mathfrak{p}$ is zero as $\mathfrak{p}$ is minimal. Hence the result follows from Lemma lemma-dualizing-artinian. $\square$